

Idris Nechnech | Curriculum Vitae

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Research Interests

I am a graduate student at Seoul National University working on post-training alignment for LLMs and reinforcement learning.

Publications

Function-Level Execution Feedback for Code Preference Optimization

Idris Nechnech et al. — *Findings of the Association for Computational Linguistics: EMNLP 2026*

Proposed a step-level preference optimization framework using execution feedback as a reward signal for code generation.

Don't Filter, Fill: Dynamic Mixed-Policy GRPO for Text-to-SQL

Idris Nechnech et al. — *Currently under review*

Proposed a mixed-policy GRPO framework that dynamically fills on-policy rollouts with teacher-generated off-policy trajectories, enabling faster convergence and higher performance than GRPO-based baselines.

Learning Rate Transfer for Hybrid Transformer-SSM Architectures

Jimin Seo et al. (incl. Idris Nechnech) — *Advances in Neural Information Processing Systems: NeurIPS 2026*

Showed that μP with AdamW achieves near-zero learning-rate transfer across widths and depths in hybrid Transformer-SSM models.

Experience

Seoul National University

Graduate Researcher

Seoul

September 2025 – February 2027

- Conducting research on NLP and post-training alignment in the Cognitive Machine Learning Lab under the supervision of Prof. Jungwoo Lee.

Institut Polytechnique de Paris

Research Intern in Statistics

Paris

June 2025 – July 2025

- Research internship at the SAMOVAR laboratory under the supervision of Prof. Stefano Fortunati.

Agence Nationale des Fréquences

Radiofrequency Technician

Paris

July 2024 – August 2024

- Monitored and controlled transmission spectrum at Paris 2024 Olympic venues to ensure interference-free communication.

Education

Seoul National University

Master of Science – MSc in Electrical and Computer Engineering

Seoul

September 2025 – February 2027

Relevant courses: Deep Learning, Natural Language Processing, Linear System Theory, Topics in Computer and VLSI

Institut Polytechnique de Paris – Télécom SudParis

Master of Engineering – MEng

Paris

September 2023 – February 2027

Ranked in the top 10% of my cohort. (GPA: 3.75/4.00)

Relevant courses: Applied Statistics, Optimization, Probability, Systems Programming, Stochastic Processes, Signal Theory.

Academic exchange at The Chinese University of Hong Kong, September 2024 – January 2025

Graduate courses: *Advanced Topics in AI, Data Science in Practice, Probabilistic Models and Inference Algorithms for Machine Learning.*

Lycée Saint-Louis

Bachelor of Science – BSc in Mathematics and Physics (PCSI/PC)

Paris

September 2021 – August 2023

Two years of intensive preparation for national competitive entrance exams to leading French "grandes écoles," specializing in mathematics, physics, and chemistry.

Projects

GenoXplain: Breast Cancer Driver Mutation Prediction

Development of an Interpretable ML Pipeline for Driver Mutation Identification

2025

- Awarded 2nd place (5,000€ prize) at the D4GEN Hackathon for an interpretable cancer genomics ML pipeline.
- Built an XGBoost-based meta-learner integrating genomic variant and structural protein features.

Research project in Statistics

Misspecified Statistics and Autoregressive Processes: An Analysis of Cramér-Rao Bounds

2025

- Conducted simulations to evaluate the performance of robust estimators (M-, R-, Newton, and ML) for AR(1) models with t-distributed noise.
- Quantified the effects of model misspecification on estimator efficiency, highlighting differences in MSE and convergence behavior.

Skills

Programming & Tools: Python, C, Java, Git, Linux, Slurm, Docker, AWS

Frameworks & Libraries: PyTorch, Hugging Face (PEFT, TRL), VERL, Ray, DeepSpeed, vLLM, Scikit-learn

Languages: French (native), English (C1: TOEFL 109/120), Korean (conversational), Japanese (conversational)